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COAL ASSIGNMENT # 1

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Q1 **a** ∴ Memory to memory transfer is not valid

mov ax, [22] or mov al, [22]
mov [02], ax mov [02], [al]

b ∴ Invalid syntax

mov al, 22
mov [wordvar], al

c ∴ Not allowed to move a 8-bit register to a 16-bit register

mov bl, al

d ∴ Addition of two index register in one memory is not allowed

mov bx, si
mov ax, [bx + di + 100]

Question 2

```
[org 0x0100]

mov al, array[0] ; removing negative
NOT al            ; sign from -3
INC al
mov array[0], al

mov al, array[2] ; removing negative
NOT al            ; sign from -10
INC al
mov array[2], al

mov al, array[5] ; removing negative
NOT al            ; sign from -7
INC al
mov array[5], al

mov bl, 0 ; initalize array index to zero
mov al, 0 ; initalize min to zero

mov al, [array+bl] ; mov number to an
mov cl, [size]

maxvalue: cmp al, [array + bl] ; find the maximum number
          jge maxloop
          mov al, [array+bl]

maxloop:  add bl, 1 ; advance bx to next index
          loop maxvalue

          mov [max], al ; write back maximum in memory
```

```
mov ah, 0x4c00 ; terminate
int 0x21        ; program

array: db -3, 5, -10, 4, 6, -7, 1
size: db 7
max: db 0
```

Q3

a Physical Address

$$
\begin{array}{r}
1\,1 \\
= 1DDD0 \\
00436 \\
\hline
\boxed{1E206}
\end{array}
$$

b Physical Address

$$
\begin{array}{r}
= 12340 \\
07920 \\
\hline
\boxed{19C60}
\end{array}
$$

c Physical Address

$$
\begin{array}{r}
1 \\
= 74F00 \\
02123 \\
\hline
\boxed{77023}
\end{array}
$$

d Physical Address

$$
\begin{array}{r}
= 00000 \\
06727 \\
\hline
\boxed{06727}
\end{array}
$$

e Physical Address

$$
\begin{array}{r}
1\;\;1\;1 \\
= FFFF0 \\
04336 \\
\hline
\boxed{\boxed{1}\,04326}
\end{array}
$$

Carry removed

f Physical Address = 10800
00100
$\boxed{10900}$

g Physical Address = AB010
0FFFF
$\boxed{BB00F}$

Q4 a	First Physical Address	Last Physical Address
a	1000	
	0x10000 +0x00000 = 0x10000	0x10000 +0x0FFFF = 0x1FFFF
b	0FFF	
	0x0FFF0 +0x00000 = 0x0FFF0	0x0FFF0 +0x0FFFF = 0x1FFEF
c	1002	
	0x10020 +0x00000 = 0x10020	0x10020 +0x0FFFF = 0x2001F
d	0001	
	0x00010 +0x00000 = 0x00010	0x00010 +0x0FFFF 0x1000F

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e E000

0x E0000	0x E0000
+ 0x 00000	+ 0x 0FFFF
= 0x E0000	0x EFFFF

Question 5

a mov ax, [bx + 12]

Effective Address = bx + 12
= 0100 + 000C
= 010C

b mov ax, [bx + num1]

Effective Address = bx + num1
= 0100 + 1001
= 1101

c mov ax, [num1 + bx]

Effective Address = num1 + bx
= 1001 + 0100
= 1101

d mov ax, [bx + si]

Effective Address = bx + si
= 0100 + 0100
= 0200

0 Question 6

a Invalid (due to base register and index of stack)

b Invalid (Base registers are not allowed)

c Effective Address = 0100
+ 0010

$\boxed{0X0110}$

d Effective Address = 0100 x 10
= 01000 - 10
= $\boxed{0X0010}$

e Invalid (Base and Stack pointer)

~~f Invalid (Base register and index)~~

f Effective Address = 0100 + 0001
$\boxed{= 0X0101}$

Question 7

a 0109

0FE20
00109
0FF29

b AX = 5

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Question 8

i   AI [1700]

$$\begin{array}{rccccc} & ^{1} & & ^{1} & & \\ & 0 & F & E & 2 & 0 \\ + & 0 & 1 & 7 & 0 & 0 \\ \hline = & 1 & 1 & 5 & 2 & 0 \end{array}$$

ii   AI [120B]

$$\begin{array}{rccccc} & ^{1} & & ^{1} & & \\ & 0 & F & E & 2 & 0 \\ + & 0 & 1 & 2 & 0 & B \\ \hline = & 1 & 1 & 0 & 2 & B \end{array}$$

Question 9

```
jmp start
~~data segment~~
multiplicand: dd 11223344H
multiplier: dd 55667788H
result: dq 0000000000000000
~~data~~
start: mov ax, data
       mov ds, ax
       Mov ax, word [multiplicand]
       mul word [multiplier]
       mov word [result], ax
       mov cx, dx
       mov ax, word [multiplicand + 2]
       mul word [multiplier]
       add cx, ax
       mov bx, dx
       jnc move
       add bx, 0001H
move: mov ax, word [multiplicand]
       mul word [multiplier + 2]
       add cx, ax
       mov word [result + 2], cx
       mov cx, dx
       jnc l1
       add bx, 0001H
l1:    mov ax, word [multiplicand + 2]
       mul word [multiplier + 2]
       add cx, ax
       jnc l2
       add dx, 0001H
l2:    add cx, bx
       mov word [result + 4], cx
```

```
    jnc l3
    add dx, 0001H
l3: mov word [result + 6], dx
 int 3
mov ax, 0x4c00
 int 0x21
```

Question 10

i) 68 bytes

ii) arr = 8, 10, 2, 0, 7, 5, 3

iii)
4) L1:
5) add si, 1
6) cmp si, di
7) je end
8) mov al, [arr + si]
9) test al, 01
10) je L1

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